

postfix.c X Siva.c QueueWA.c X Circular.c X

Siva.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int n , i , flag ;
5      int low, high, mid;
6      int comparisons = 0;
7      int found = 0;
8      printf("Enter number of employee IDs: ");
9      scanf("%d", &n);
10     int id[n];
11     printf("Enter %d employee IDs in ascending order:\n", n);
12
13     for (i = 0; i < n; i++)
14     {
15         scanf("%d", &id[i]);
16     }
17
18     printf("Enter employee ID to search: ");
19     scanf("%d", &flag);
20     low = 0;
21     high = n - 1;
22     while (low <= high)
23     {
24         mid = (low + high) / 2;
25         comparisons++;
26
27         if (id[mid] == flag)
28         {
29             found = 1;
30             break;
31         }
32         else if (flag < id[mid])
```

```
9     else if (flag < id[mid])
10     {
11         high = mid - 1;
12     }
13     else
14     {
15         low = mid + 1;
16     }
17 }
18
19 if (found==1)
20 {
21     printf("Employee ID %d found at position %d\n",
22         flag, mid + 1);
23 }
24 else
25 {
26     printf("Employee ID %d is absent\n", flag);
27 }
28
29 printf("Number of comparisons = %d\n", comparisons);
30 return 0;
31 }
```

```
PS C:\Users\USER\OneDrive\Desktop\DS with c> gcc Siva.c -o Siva
```

```
PS C:\Users\USER\OneDrive\Desktop\DS with c> .\Siva.exe
```

```
Enter number of employee IDs: 7
```

```
Enter 7 employee IDs in ascending order:
```

```
100
```

```
105
```

```
110
```

```
115
```

```
120
```

```
125
```

```
130
```

```
Enter employee ID to search: 120
```

```
Employee ID 120 found at position 5
```

```
Number of comparisons = 3
```

```
PS C:\Users\USER\OneDrive\Desktop\DS with c> .\Siva.exe
```

```
Enter number of employee IDs: 7
```

```
Enter 7 employee IDs in ascending order:
```

```
100
```

```
105
```

```
110
```

```
115
```

```
120
```

```
125
```

```
130
```

```
Enter employee ID to search: 118
```

```
Employee ID 118 is absent
```

```
Number of comparisons = 3
```

```
PS C:\Users\USER\OneDrive\Desktop\DS with c> 
```